

TradeAlerts Project Description

Overview

TradeAlerts is a Flask-based web application for live and simulated stock trading signals. The system now includes **real-time dashboard alerts**, an advanced **backtesting engine with full parameter control and performance visualization**, and seamless workflow for deploying backtested strategies to production.

Major Features

- **Dashboard**

- Real-time alerts (Prime/Sell) with filter, sparklines, triggers, and news links.

- Table with confidence, price, timestamp, trigger badges, and buy/sell actions.

- Persistent nav bar for one-click access to all app sections.

- **Scanner**

- Loops through S&P500, computes all indicators and triggers.

- Posts new Prime alerts to the dashboard API if *all* required signals align (configurable).

- **Backtest Engine**

- Run historical simulations using your exact “Prime” logic and tweakable parameters.

- Simulate auto-buy/sell with customizable profit/stop/hold/exits.

- See summary stats (win %, avg win/loss, max drawdown), full trade log, and interactive equity/P&L charts.

- Download results as CSV.

- “Promote to Live” button applies backtest settings to your live scanner instantly.

- **Parameter Controls**

- All key strategy variables (targets, stops, hold time, required triggers) are adjustable from the UI—both for backtest and (optionally) live trading.

- Config is stored in `config.json` and read by both modules.

Core Components

- **dashboard.py**: Main Flask endpoint

- **api.py**: API blueprint (alerts, promote settings, sparkline SVG)

- **backtest_service.py**: Historical engine (simulates signals, returns stats/results)

- **scanner.py**: Real-time signal generator

- **services/**: All indicator, DB, and news logic

- **templates/**: UI for alerts, simulation, and backtest results

Usage

- Use backtest page to optimize parameters/logic (UI controls and charts)

- When ready, click “Promote to Live” to push config to production scanner

- All config changes are live instantly—no code edit required

Docs

- Project conventions, ToDo roadmap, and architecture notes are in `/docs`

Full Scanner Simulation

Our full scanner simulation engine iterates through the entire symbol universe, applies all enabled indicator filters (SMA, RSI, MACD histogram, Bollinger Bands breakout, VWAP threshold, etc.), and simulates trades in real-time or historical replay mode. It logs buy/sell events, reasons for decisions, and portfolio

performance over time.

Full Scanner Backtesting

The backtesting module replays historical 1-minute and 1-day bar data across the full symbol set, applying the same indicator logic and trade execution rules. It calculates detailed metrics including win rate, average return, maximum drawdown, and Sharpe ratio—enabling side-by-side comparison of strategy variations.

Complex Indicator Logic

We support a full suite of technical indicators:

- **Simple Moving Average (SMA):** Configurable lookback and cross-over triggers.
- **Relative Strength Index (RSI):** Overbought/oversold thresholds and slope detection.
- **MACD Histogram:** Fast/slow EMA spans and signal line for momentum.
- **Bollinger Bands:** Customizable length and standard-deviation multipliers for breakout detection.
- **Volume Weighted Average Price (VWAP):** Intraday benchmark with threshold offsets.
- **Custom Combinations:** Flexible toggles for mixing any subset of the above filters in any order.